Ferroelectric Vortex CPU Platform

Complete Engineering Specification

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11 September 2026

Document Type: Engineering Design — Core Specification
Document License: CC BY 4.0
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Technology Patent Status: **PUBLIC RELEASE — NO PATENT SOUGHT**
Advanced Optimization: *Available later in this document*
Repository: Mega — Proton Drive — Zenodo — GitHub — OSF — Academia — X

This document provides the complete engineering specification for a general-purpose processor platform built on ferroelectric vortex memory cells. The platform consists of two dies in a single package: a Main Processor with a stored-function ALU array and pipeline, and a Vector Co-Processor with a 1024×1024 crossbar and integrated non-volatile memory. All cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria are provided explicitly. Every parameter required to build, test, and operate the baseline platform is included. The design is released as a public, patent-free engineering specification. An advanced optimization configuration is offered separately and is described at the end of this document.

This is a 6DUFT inspired technology.

Abstract

This document is the complete engineering specification for a general-purpose processor platform built on ferroelectric vortex memory cells.

The platform consists of two dies in a single package:

1. **Main Processor:** A scalar processor with a stored-function ALU array, a pipeline, a cache hierarchy, and a register file.
2. **Vector Co-Processor:** A 1024×1024 crossbar for matrix-vector multiplication, with 0.109 GB of integrated non-volatile memory.

The baseline platform provides:

- **Clock:** 8.10 GHz
- **Effective IPC:** 2.614
- **Single-thread throughput:** 21.2 GIPS
- **Vector throughput:** 1.02×10^{14} MAC/s (1-bit workloads)
- **Power:** 0.90 W (Main: 221 mW, Vector Co-Processor: 680 mW)
- **Efficiency:** 23.6 GIPS/W
- **Cost:** $7.24

The platform uses standard 28 nm CMOS with hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition. No exotic materials are required.

The document covers architecture, cell geometry and materials, fabrication flows for both dies, circuit design, operating principles, performance specifications, verification and qualification, failure modes and mitigations, packaging, and an advanced optimization configuration available separately.

The design is released openly. No patent is sought.

This is a 6DUFT inspired technology.

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Part I
Introduction and Overview

1 Purpose and Scope

This document is the complete engineering specification for a general-purpose processor platform built on ferroelectric vortex memory cells.

It provides every parameter, process step, circuit specification, and procedure required to build, test, and operate the baseline platform.

The document is written for:

- Foundry process engineers who will fabricate the two dies.
- Circuit designers who will integrate the platform.
- Test engineers who will qualify the platform.
- System engineers who will deploy the platform.

No prior knowledge of any theoretical framework is required. All necessary engineering information is contained in this document.

2 Platform Overview

The platform consists of two dies in a single package:

Die	Function	Die Area
Main Processor	Scalar processor with stored-function ALU array and pipeline	65 mm^2
Vector Co-Processor	1024×1024 crossbar with 0.109 GB integrated memory	54 mm^2
Total	**Two-die package**	**119 mm^2**

2.1 Main Processor

The Main Processor is a scalar processor that:

- Runs general-purpose code.
- Executes the operating system.
- Handles branches, caches, and control flow.
- Is optimized for single-thread performance.

The Main Processor uses a stored-function ALU array instead of a traditional gate-based ALU. Each ALU is a 16×4 crossbar that stores a truth table implementing a specific 4-bit to 4-bit function. The truth table is programmed at fabrication.

2.2 Vector Co-Processor

The Vector Co-Processor is a dedicated accelerator that:

- Accelerates vector and matrix operations.
- Provides 0.109 GB of integrated non-volatile memory.
- Handles DMA transfers.
- Is fixed at the baseline configuration.

The Vector Co-Processor uses a 1024×1024 crossbar of ferroelectric cells. The crossbar stores matrix weights and performs a matrix-vector multiplication in a single parallel operation.

3 Why Two Dies

A single die would combine the scalar and vector functionality. Two dies are used instead because:

1. The Main Processor uses the ferroelectric material for the ALU array.
2. The Vector Co-Processor uses the ferroelectric material for the crossbar and memory.
3. The two dies have different array layouts and different circuit requirements.
4. Separate dies allow the platform to be upgraded by replacing only the Main Processor.
5. Yield is better with two smaller dies than one large die.

4 What the Platform Is Not

The platform is not:

- A multi-core server processor. The platform has one Main Processor core. It does not have multiple cores for multi-threaded workloads.
- A training accelerator. The Vector Co-Processor is for inference and matrix operations, not neural network training.
- A complete system-on-chip. It requires external power management, clock reference, and I/O peripherals.
- Field-programmable. The ALU truth tables are fixed at fabrication. The instruction set is fixed.

5 Key Advantages

- **Single-thread performance:** 21.2 GIPS at the baseline configuration.
- **Vector throughput:** 1.02×10^{14} MAC/s (1-bit workloads).
- **Low power:** 0.90 W total, which is 139× lower than a typical desktop processor.
- **Low cost:** $7.24 per chip.
- **Non-volatile memory:** 0.109 GB integrated, with instant boot.
- **CMOS compatible:** Standard 28 nm CMOS with HZO atomic layer deposition.
- **No exotic materials.**

6 Key Limitations

- **Baseline single-thread throughput:** 0.91× a current flagship desktop processor.
- **Single-core:** No multi-core scaling.
- **Integrated memory capacity:** 0.109 GB.
- **Fixed function:** The ALU truth tables are programmed at fabrication.

7 Platform Positioning

The baseline platform is positioned for:

- Embedded and edge applications
- Low-power desktop and mobile systems
- Vector-heavy workloads
- Educational and research use

An advanced optimization configuration with higher clock speed, higher throughput, and additional capability is available separately and is described at the end of this document.

Part II
Device Architecture

8 Main Processor Architecture

8.1 Overview

The Main Processor consists of:

- A stored-function ALU array (1000 ALUs, each a 16×4 crossbar).
- A pipelined execution core.
- A register file.
- A branch predictor.
- An L1 cache hierarchy (64 KB instruction, 64 KB data).
- An L2 unified cache (1 MB).
- Control logic.
- Clock distribution network.

8.2 ALU Array

Each ALU is a 16×4 crossbar of ferroelectric cells. The crossbar stores a truth table that implements the ALU function.

Parameter	Value
Rows per ALU	16
Columns per ALU	4
Cells per ALU	64
Total ALUs	1000
Total ALU cells	64,000
Function selection	Instruction opcode

The ALU operation is a single read:

1. The 4-bit input selects one of 16 rows.
2. The selected row applies voltage to the word line.
3. The crossbar computes the 4 output bits in parallel.
4. The comparator reads each output column.
5. The 4-bit result is latched.

Each ALU implements one specific 4-bit to 4-bit function. The function is selected by the instruction decoder. Since there are only 256 possible 4-bit to 4-bit functions, and 1000 ALUs available, each function is implemented approximately 4 times. This gives redundancy.

8.3 Pipeline

The pipeline has 14 stages:

Stage	Function	Sub-Stages
1–5	Instruction fetch (L1I)	5
6–7	Decode	2
8–10	Operand fetch	3
11	Execute	1
12	Writeback	1
13	Branch resolution	1
14	Retire	1
Total		**14**

8.4 Register File

Parameter	Value
Scalar registers	32
Register width	64 bits
Read ports	8
Write ports	4
Access time	100 ps
Area	0.001 mm^2

8.5 Branch Predictor

Parameter	Value
Type	Tournament (bimodal + gshare)
Bimodal entries	1024
Gshare entries	1024
Selector entries	1024
Prediction accuracy	95%
Misprediction penalty	14 cycles

8.6 Cache Hierarchy

Cache	Size	Ways	Access
L1 Instruction	64 KB	4	500 ps
L1 Data	64 KB	4	500 ps
L2 Unified	1 MB	8	2 ns
Vector Co-Processor memory	0.109 GB	—	3.74 ns

9 Vector Co-Processor Architecture

9.1 Crossbar

Parameter	Value
Crossbar size	1024×1024
Cell material	$Hf_{0.5}Zr_{0.5}O_2$
Cell type	200 nm cell, 100 nm disk, 20 nm pillar
Cell film thickness	14.83 nm
Compute time	10.3 ns

9.2 Peripheral Circuits

The Vector Co-Processor includes:

- 1024 DACs (4-bit, one per word line)
- 1024 TIAs (transimpedance amplifiers)
- 1024 ADCs (8-bit, one per bit line)
- 1024 memory comparators
- Control logic

9.3 Integrated Memory

Parameter	Value
Capacity	0.109 GB
Cell material	$Hf_{0.5}Zr_{0.5}O_2$
Read latency	3.74 ns
Write latency	21.1 ns
Power (read)	0.847 mW (comparators)

10 Inter-Die Interface

The two dies communicate through the package substrate.

Parameter	Value
Traces	464 (232 signal + 232 ground)
Trace pitch	10 μm
Data rate	10 Gb/s per trace
Total bandwidth	290 GB/s

11 Clock Domains

The platform uses two clock domains on each die.

Domain	Frequency	Used by
Core logic	500 MHz	Control, cache, branch predictor, I/O
Compute	8.10 GHz (baseline)	ALU array, word line drivers, comparators

The two clock domains are synchronized through an asynchronous FIFO at the boundary between the core logic and the ALU array.

Part III
Cell Geometry and Materials

12 Main Processor ALU Cell

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Cell size	100 nm × 100 nm
Disk diameter	60 nm
Pillar diameter	40 nm
Pillar height	4.90 nm
Film thickness	14.83 nm
Annular gap (pillar to disk edge)	10 nm
Resistance ($n = 1$)	6,202 Ω
Capacitance	61.74 aF
Settling time	0.383 ps
Read voltage	9.886 mV
Read current	1.373 μA

13 Vector Co-Processor Cell

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Cell size	200 nm × 200 nm
Disk diameter	100 nm
Pillar diameter	20 nm
Pillar height	2 nm
Film thickness	14.83 nm
Annular gap (pillar to disk edge)	40 nm
Annular area	161.9 nm^2
Resistance ($n = 1$)	11,762 Ω
Capacitance	23.38 aF
Settling time	0.275 ps

14 Material Properties

14.1 Hafnium Zirconium Oxide

Property	Value
Composition	$Hf_{0.5}Zr_{0.5}O_2$
Film thickness (Main Processor)	14.83 nm
Film thickness (Vector Co-Processor)	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m

14.2 Electrode Materials

Electrode	Material	Thickness
Bottom	TiN	50 nm
Top	TiN	50 nm
Pillar (Main Processor)	TiN	4.90 nm
Pillar (Vector Co-Processor)	TiN	2 nm
Etch stop	Al_2O_3	2 nm

Part IV
Fabrication Flow

15 Overview

Two separate fabrication flows are used:

- **Main Processor:** 19 steps, 10 masks.
- **Vector Co-Processor:** 18 steps, 10 masks.

The Main Processor and Vector Co-Processor are fabricated on separate wafers and then packaged together.

16 Main Processor Fabrication Flow

16.1 Step 1: Substrate Preparation

Parameter	Value
Wafer size	300 mm
Substrate	Silicon (100)
Thermal SiO_2	100 nm ± 2 nm

16.2 Step 2: FinFET Fabrication

Standard 28 nm CMOS front-end.

Parameter	Value
Gate length	28 nm
Fins	3-fin
Gate oxide	High-κ (HfO_2, EOT 3 nm)
Source/drain	$CoSi_2$
Threshold voltage	0.3 V

16.3 Step 3: Bottom Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C

16.4 Step 4: Etch Stop Deposition

Parameter	Value
Material	Al_2O_3
Thickness	2 nm ± 0.2 nm
Method	ALD

16.5 Step 5: Pillar Layer Deposition

Parameter	Value
Material	TiN
Thickness	4.90 nm
Method	ALD

16.6 Step 6: Pillar Patterning

Parameter	Value
Diameter	40 nm
Method	Immersion lithography (193 nm)
Alignment tolerance	±2 nm

16.7 Step 7: Pillar Etch

Parameter	Value
Chemistry	Cl_2/Ar
Method	RIE
Etch stop	Al_2O_3 layer
Overetch	10%

16.8 Step 8: Etch Stop Removal

Parameter	Value
Etchant	Dilute HF (0.1%)
Etch time	4 s

16.9 Step 9: HZO Deposition

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Thickness	14.83 nm
Method	ALD
Temperature	250°C
Supercycles	82
Deposition time	17 min

Process note: The film must be annealed within 1 hour of deposition.

16.10 Step 10: RTA Annealing

Parameter	Value
Temperature	450°C
Duration	20 s
Atmosphere	N_2

16.11 Step 11: Top Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm
Method	ALD

16.12 Step 12: Top Electrode Patterning

Parameter	Value
Pattern	Circle
Diameter	60 nm
Alignment	Centered on pillar

16.13 Step 13: Disk Etch

Parameter	Value
Stack	TiN (50 nm) + HZO (14.83 nm) + TiN (50 nm)
Chemistry (TiN)	Cl_2/Ar
Chemistry (HZO)	BCl_3/Ar
Etch stop	SiO_2 buried oxide

16.14 Step 14: Post-Etch Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	O_2

16.15 Step 15: Passivation

Parameter	Value
Material	SiN (50 nm) + SiO_2 (50 nm)
Method	PECVD
Temperature	350°C

16.16 Step 16: Contact Via Formation

Parameter	Value
Via diameter	50 nm
Fill material	Tungsten (CVD)

16.17 Step 17: Cu Interconnect

Parameter	Value
Thickness	1000 nm
Barrier	Ta/TaN (10 nm)
Fill	Cu electroplating
CMP	After fill

16.18 Step 18: Thick Metal Layers

The Main Processor uses thick metal layers for power delivery to the ALU array.

Parameter	Value
Layers	2 (M9 and M10)
Thickness	10 μm each
Material	Cu
Purpose	Power distribution
Sheet resistance	2.0×10^{-3} Ω/square
Total power-delivery resistance	0.1 mΩ

The thick metal layers reduce the power-delivery resistance and permit the full active ALU count to be supplied without IR-drop penalty.

16.19 Step 19: Final Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)

16.20 Main Processor Mask Set

Mask	Purpose	CD
M1	FinFET gate	28 nm
M2	Bottom electrode	100 nm
M3	Pillar pattern	40 nm
M4	Top electrode	60 nm
M5	Disk pattern	60 nm
M6	Contact vias	50 nm
M7	Cu interconnect	1000 nm
M8	Thick metal	10 μm
M9	Bond pad opening	50 μm
M10	Scribe line test structures	—

17 Vector Co-Processor Fabrication Flow

The Vector Co-Processor uses the same 18-step flow as the Memory Device. The full flow is included here for completeness.

17.1 Steps 1–18

Same as the Memory Device fabrication flow:

- Step 1: Substrate preparation
- Step 2: FinFET fabrication (3-fin)
- Step 3: Bottom electrode deposition (TiN, 50 nm)
- Step 4: Etch stop deposition (Al_2O_3, 2 nm)
- Step 5: Pillar layer deposition (TiN, 2 nm)
- Step 6: Pillar patterning (20 nm)
- Step 7: Pillar etch
- Step 8: Etch stop removal
- Step 9: HZO deposition (14.83 nm, 82 supercycles)
- Step 10: RTA annealing (450°C, 20 s)
- Step 11: Top electrode deposition (TiN, 50 nm)
- Step 12: Top electrode patterning (100 nm)
- Step 13: Disk etch

- Step 14: Post-etch anneal (400°C, 30 min)
- Step 15: Passivation
- Step 16: Contact via formation
- Step 17: Cu interconnect
- Step 18: Final anneal

17.2 Vector Co-Processor Mask Set

Mask	Purpose	CD
M1	FinFET gate	28 nm
M2	Bottom electrode	100 nm
M3	Pillar pattern	20 nm
M4	Top electrode	100 nm
M5	Disk pattern	100 nm
M6	Contact vias	50 nm
M7	Cu interconnect	1000 nm
M8	Bond pad opening	50 μm
M9	Passivation via	50 nm
M10	Scribe line test structures	—

18 Metal Stack

The Main Processor uses a 10-layer copper metal stack.

Layer	Thickness	Pitch	Function
M1	100 nm	100 nm	Local interconnect
M2–M5	200 nm each	200 nm	Signal routing
M6–M7	400 nm each	400 nm	Signal routing, clock
M8	1000 nm	1 μm	Power distribution
M9	10 μm	10 μm	Power distribution (thick)
M10	10 μm	10 μm	Power distribution (thick)

Dielectric stack: standard low-κ (SiOCH) inter-metal dielectric.
The Vector Co-Processor uses a 7-layer copper metal stack.

Layer	Thickness	Pitch	Function
M1	100 nm	100 nm	Local interconnect
M2–M3	200 nm each	200 nm	Signal routing
M4–M5	400 nm each	400 nm	Signal routing, clock
M6	1000 nm	1 μm	Power distribution
M7	2000 nm	2 μm	Power distribution

19 Yield Analysis

19.1 Main Processor

Parameter	Value
Die area	65 mm^2
Defect density	0.5 cm^{-2}
Yield	72.3%
Good dies per 300 mm wafer	754

19.2 Vector Co-Processor

Parameter	Value
Die area	54 mm^2
Defect density	0.5 cm^{-2}
Yield	76.3%
Good dies per 300 mm wafer	959

20 Cost Analysis

Component	Cost	Notes
Main Processor die	\$2.65	
Vector Co-Processor die	\$2.09	
Package	\$2.50	23 mm $\times$ 23 mm BGA
Total chip cost	**\$7.24**	

Part V
Circuit Design

21 Main Processor Circuits

21.1 Comparator

Parameter	Value
Type	Current-mode comparator
Topology	Double-tail latch with pre-amplification
Decision time	122.2 ps
Supply voltage	1.0 V
Power per comparator	1.373 μW
Number	4,000

21.2 Word Line Driver

Parameter	Value
Type	CMOS buffer
Output voltage	0–1 V
Rise time	2.25 ps
Drive capability	299 μA
Number	16,000

21.3 Write Driver

Parameter	Value
Topology	H-bridge
Voltage range	-3.11 V to $+3.11$ V
Transistor type	5 V I/O

21.4 Charge Pump

Parameter	Value
Topology	Dickson 5-stage
Input	+1 V
Output	-3.5 V
Efficiency	80%

21.5 Register File, Branch Predictor, Cache

Parameter	Value
Register file	32 registers × 64 bits, 8 read / 4 write ports
Branch predictor	Tournament, 1024 entries, 95% accuracy
L1 cache	64 KB instruction + 64 KB data, 4-way
L2 cache	1 MB unified, 8-way

21.6 Clock Generation

Parameter	Value
Reference	100 MHz
PLL	Charge-pump PLL
Multiplier	81×
Jitter	<1 ps RMS
Lock time	<10 μs

21.7 Main Processor Power Breakdown

Component	Power
ALU comparators	22.0 μW
Clock distribution	68 mW
Control	17 mW
I/O	100 mW
Cache	17 mW
ECC	10 mW
Leakage	8.75 mW
Total	**221 mW**

22 Vector Co-Processor Circuits

22.1 Comparator Array

Parameter	Value
Decision time	199.5 ps
Power per comparator	0.840 μW
Number	1024

22.2 DAC Array

Parameter	Value
Resolution	4 bits
Full scale	0–50 mV
Settling time	1 ns
Power per DAC	100 μW
Number	1024

22.3 TIA Array

Parameter	Value
Transimpedance	10 kΩ
Bandwidth	1 GHz
Settling time	0.265 ns
Power per TIA	50 μW
Number	1024

22.4 ADC Array

Parameter	Value
Resolution	8 bits
Sample rate	100 MS/s
Conversion time	10 ns
Power per ADC	469 μW
Number	1024

22.5 Vector Co-Processor Power Breakdown

Component	Power
Memory comparators	0.847 mW
DACs	102.4 mW
ADCs	480.3 mW
TIAs	51.2 mW
Control	20 mW
I/O	50 mW
Clock	10 mW
ECC	10 mW
Leakage	6 mW
Total	**680 mW**

23 Inter-Die Interface

Parameter	Value
Traces	464 (232 signal + 232 ground)
Data rate	10 Gb/s per trace
Total bandwidth	290 GB/s
Driver type	LVDS
Receiver type	Differential amplifier
Power (TX)	20 mW
Power (RX)	15 mW

24 Design for Test

- JTAG test access port.
- Built-in self-test (BIST) with memory exerciser.

- Redundancy: 2 spare rows and 2 spare columns per sub-array.
- Laser fuse programming for repair.

Part VI
Operating Principles

25 Instruction Execution

25.1 Instruction Procedure

1. Instruction fetch (L1I cache access).
2. Instruction decode (opcode, operands).
3. Operand fetch (register file read).
4. Execute (ALU truth table read).
5. Memory access (L1D cache, if needed).
6. Writeback (register file write).
7. Branch resolution (if branch).
8. Retire.

25.2 Execute Timing

Component	Time
Settling	0.383 ps
Comparator	122.2 ps
Overhead	0.94 ps
Total	**123.5 ps**

25.3 Clock Frequency

$$f = \frac{1}{123.5 \text{ ps}} = 8.10 \text{ GHz} \tag{1}$$

26 Vector Compute Operation

26.1 Procedure

1. Load input vector to DACs (1 ns settling).
2. Apply voltages to word lines (0.034 ns).
3. Current summing on bit lines.
4. TIA converts current to voltage (0.265 ns).
5. ADC digitizes (10 ns).
6. Output vector read.

26.2 Compute Time

$$t_{\text{compute}} = 0.034 + 0.265 + 10 = 10.3 \text{ ns} \tag{2}$$

26.3 Vector Throughput

$$\text{Throughput} = \frac{1{,}048{,}576}{10.3 \times 10^{-9}} = 1.02 \times 10^{14} \text{ MAC/s} \tag{3}$$

27 Programmable Function Selection

Each ALU implements one specific 4-bit to 4-bit function. The function is selected by the instruction decoder.

Since there are 256 possible 4-bit to 4-bit functions, and 1000 ALUs are available, each function is implemented approximately 4 times. The instruction decoder selects the appropriate ALU for each opcode.

28 Write Operation (Programming at Fabrication)

The ALU truth tables are programmed at fabrication during production test. The programming occurs only once. The truth tables cannot be changed in the field.

28.1 Write-Verify Voltage Levels

Attempt	Field	Voltage	Cumulative
1	1.5 MV/cm	2.2245 V	0.500
2	1.7 MV/cm	2.5211 V	0.875
3	1.9 MV/cm	2.8177 V	0.984
4	2.1 MV/cm	3.1143 V	0.999

28.2 Programming Time

With 100 cells programmed in parallel:

$$t_{\text{program}} = \frac{64{,}000 \times 22.2 \text{ ns}}{100} = 14.2 \ \mu\text{s} \tag{4}$$

29 Power-On Sequence

1. Apply 1 V logic supply.
2. Wait for power-on reset (1 μs).
3. Apply write voltage supply (only during programming).
4. Load calibration from on-chip memory.
5. Begin operation.

Total boot time: $<$10 μs.

30 Power-Down Sequence

1. Complete pending operations.
2. Store critical data to non-volatile memory.
3. Remove write supply.
4. Remove logic supply.

Non-volatile memory preserves all state. Power-off is instant.

31 Error Handling

31.1 Correctable Errors

1. Correct the error using ECC.
2. Log the event.
3. Continue operation.

31.2 Uncorrectable Errors

1. Retry the operation.
2. If retry fails, mark block as bad.
3. Use a spare block.
4. If no spares remain, report to OS.

Part VII
Performance Specifications

32 Summary of Baseline Performance

Parameter	Value
Clock	8.10 GHz
Effective IPC	2.614
Single-thread throughput	21.2 GIPS
Vector throughput (1-bit)	1.02×10^{14} MAC/s
Vector throughput (INT8-equiv)	1.28×10^{13} MAC/s
Main Processor power	221 mW
Vector Co-Processor power	680 mW
Total power	0.90 W
Efficiency	23.6 GIPS/W
Die area (total)	119 mm^2 (two dies)
Chip cost	$7.24
Pipeline depth	14 stages
Supply voltage	1.0 V logic

33 Comparison with Intel i9-13900K

Parameter	Intel i9	This Platform
Single-thread	23.2 GIPS	21.2 GIPS
Multi-thread	557 GIPS	21.2 GIPS
Vector	2.97×10^{12} MAC/s	1.02×10^{14} MAC/s
Power	125 W	0.90 W
Die area	257 mm^2	119 mm^2 (two dies)
Chip cost	\$600	\$7.24

34 Honest Assessment

34.1 Strengths

- Vector throughput is 34.3× faster than Intel AVX-512 for 1-bit workloads.
- Power is 139× lower than a typical desktop processor.
- Cost is 83× lower than a typical desktop processor.
- Efficiency is 127× better than a typical desktop processor.
- Non-volatile memory with instant boot.

34.2 Weaknesses

- Single-thread throughput is 0.91× a current flagship desktop processor.
- Single-core: no multi-core scaling.
- Integrated memory capacity is 0.109 GB.
- Fixed function: the ALU truth tables are programmed at fabrication.

Part VIII
Verification and Qualification

35 Overview

The verification plan follows the three-gate structure:

1. **Gate 1:** The ferroelectric phase is confirmed immediately after annealing.
2. **Gate 2:** The ALU cell switches correctly at the single-cell level.
3. **Gate 3:** The device works at scale in an array.

36 Gate 1: Ferroelectric Phase Confirmation

36.1 When to Measure

After Step 10 (RTA Annealing).

36.2 What to Measure

A P-V hysteresis loop on a TiN/HZO/TiN capacitor.

36.3 Pass Criteria

Measurement	Target	Pass
Hysteresis loop	Open loop	Clearly visible
Remanent polarization	>10 μC/cm^2	Above threshold
Coercive field	1–2 MV/cm	Within range
Film thickness	14.83 nm $\pm$ 1%	Ellipsometry

37 Gate 2: Single Cell Verification

37.1 Procedure

1. Probe the top and bottom electrodes.
2. Apply the write voltage (2.2245 V) for 10 ns.
3. Apply the read voltage (9.886 mV) and measure the current.
4. Apply the erase voltage for 10 ns.
5. Apply the read voltage and measure the current.
6. Repeat 10 times.

37.2 Pass Criteria

Measurement	Target	Pass
$n = 1$ read current	1.373 μA (Main), 840.5 nA (Vector)	Within 50%
$n = 0$ read current	<1 pA	Below threshold
ON/OFF ratio	$>10^9$	Above threshold
Reproducibility	10/10 cycles	All succeed

38 Gate 3: Small Array Verification

38.1 Procedure

1. Write all 4,096 cells in a 64$\times$64 test array to $n = 1$.
2. Read all cells and measure current.
3. Write all cells to $n = 0$.
4. Read all cells and measure current.
5. Calculate yield and cell-to-cell variation.

38.2 Pass Criteria

Measurement	Target	Pass
Array yield	>80%	Above threshold
Cell variation	<20% (1σ)	Below threshold
Write/read success	100%	No systematic failures

39 Reliability Testing

Test	Specification
HTOL	1000h at 125°C
HTSL	1000h at 150°C
Temperature cycling	500 cycles, −40 to 125°C
HAST	96h at 130°C/85% RH
ESD	HBM 2 kV, CDM 500 V
Latchup	100 mA, 85°C
Endurance	10^9 cycles at 85°C
Retention	1000h at 85°C

40 Burn-In

Parameter	Value
Temperature	125°C
Voltage	1.1× nominal
Duration	48 hours
Pattern	Processor exerciser + memory test

Part IX
Failure Modes and Mitigations

41 Material Failure Modes

Mode	Cause	Severity	Mitigation
HZO not ferroelectric	Wrong anneal temperature	HIGH	Increase anneal to 500°C
HZO phase degradation	Grain growth	MEDIUM	Control RTA profile
Pinholes in HZO	Contamination	HIGH	Cleanroom, ALD QC
Film thickness error	ALD cycle count error	HIGH	Ellipsometry verification
Moisture absorption	Delay before anneal	HIGH	Anneal within 1 hour
Film thickness variation	ALD non-uniformity	MEDIUM	Ellipsometry at 9 sites

42 Device Failure Modes

Mode	Cause	Severity	Mitigation
Pillar over-etch	No etch stop	HIGH	Al_2O_3 etch stop layer
Disk over-etch	Poor selectivity	HIGH	Endpoint detection
Alignment error	Lithography drift	MEDIUM	± 2 nm overlay tolerance
Vortex not forming	Annular width too small	HIGH	Verify annular width is adequate for vortex formation

43 Circuit Failure Modes

Mode	Cause	Severity	Mitigation
Comparator offset	Transistor mismatch	MEDIUM	Auto-zeroing
Register file corruption	Write port conflict	HIGH	Scoreboard, operand forwarding
Branch predictor failure	Pattern aliasing	LOW	Increase table size
Cache coherence error	Protocol bug	HIGH	Formal verification
Clock jitter	PLL noise	LOW	<1 ps RMS specification

44 System Failure Modes

Mode	Cause	Severity	Mitigation
Cell fatigue	Endurance exceeded	MEDIUM	Write-once for ALU cells
Weight drift	Retention failure	MEDIUM	Periodic recalibration
Thermal runaway	Cooling failure	HIGH	Thermal shutdown
Read disturb	Read voltage too high	HIGH	Verify margin
Package warpage	Stress imbalance	MEDIUM	Stress compensation

Mode	Cause	Severity	Mitigation
Inter-die failure	Trace breakage	HIGH	Redundant traces

Part X
Packaging

45 Package Specification

Parameter	Value
Package type	BGA
Ball count	484
Ball pitch	0.8 mm
Package size	23 mm × 23 mm
Package height	1.2 mm
Thermal resistance	15°C/W
Die attach	Epoxy, 25 μm
Wire type	Gold, 25 μm
Bond pad pitch	100 μm

46 Pad Ring Layout

The pad ring uses the foundry's standard pad ring generator.

Parameter	Value
Pad ring style	Staggered
Pad pitch	100 μm
Pad opening	50 μm × 50 μm
Pad type	Standard foundry I/O cells
Power pads	Distributed around perimeter
Signal pads	Ordered by function

The pad ring is generated from the ball assignment using the foundry's standard pad ring tool.

47 Thermal Analysis

Parameter	Value	Notes
Total power	0.90 W	Main + Vector
Thermal resistance	15°C/W	Package
Temperature rise	13.5°C	Above ambient
Junction temperature	38.5°C	At 25°C ambient

All operating temperatures remain well within the 0–85°C range. No active cooling is required.

Part XI
Advanced Optimization Available

ADVANCED OPTIMIZATION AVAILABLE

The design specified in this document is the baseline implementation of the ferroelectric vortex CPU platform.
Further optimization is possible.
The advanced optimization provides:

- **Clock:** 16 GHz (1.98× the baseline)
- **Effective IPC:** 2.484
- **Single-thread throughput:** 39.7 GIPS (1.87× the baseline)
- **Vector throughput:** 1.02×10^{14} MAC/s (unchanged)
- **Power:** 1.10 W (Main: 419 mW, Vector Co-Processor: 680 mW)
- **Efficiency:** 36.1 GIPS/W (1.53× the baseline)
- **Cost:** $7.34

The complete specifications for the advanced optimization are available from the author.
Contact: M.V.Hristov@proton.me
The advanced optimization is not included in this document.

Part XII
Appendix A: Complete Numerical Tables

A Main Processor Parameters

Parameter	Value
Clock	8.10 GHz
Effective IPC	2.614
Single-thread throughput	21.2 GIPS
Pipeline depth	14 stages
ALU count	1000
ALU size	16×4
Cell size	100 nm
Disk diameter	60 nm
Pillar diameter	40 nm
Pillar height	4.90 nm
Film thickness	14.83 nm
Cell resistance ($n = 1$)	6,202 Ω

Parameter	Value
Comparator decision time	122.2 ps
Read voltage	9.886 mV
Read current	1.373 μA
Power	221 mW
Die area	65 mm^2

B Vector Co-Processor Parameters

Parameter	Value
Crossbar size	1024×1024
Compute time	10.3 ns
Vector throughput (1-bit)	1.02×10^{14} MAC/s
Vector throughput (INT8-equiv)	1.28×10^{13} MAC/s
Cell size	200 nm
Disk diameter	100 nm
Pillar diameter	20 nm
Pillar height	2 nm
Film thickness	14.83 nm
Cell resistance ($n = 1$)	11,762 Ω
Integrated memory	0.109 GB
Memory read latency	3.74 ns
Power	680 mW
Die area	54 mm^2

C Inter-Die Interface Parameters

Parameter	Value
Traces	464 (232 signal + 232 ground)
Trace pitch	10 μm
Data rate	10 Gb/s per trace
Total bandwidth	290 GB/s

D Complete Platform Parameters

Parameter	Value
Total power	0.90 W

Parameter	Value
Total die area	119 mm^2
Chip cost	$7.24
Operating temperature	0–85°C
Supply voltage	1.0 V logic

E Fabrication Summary

Die	Steps	Masks	Notes
Main Processor	19	10	Thick metal layers included
Vector Co-Processor	18	10	Standard HZO process

Part XIII
Appendix B: Glossary

Term	Definition
Stored-function ALU	An ALU implemented as a 16×4 crossbar that stores a truth table for a specific 4-bit to 4-bit function
Ferroelectric cell	A memory cell that stores information in the state of a ferroelectric material
HZO	Hafnium zirconium oxide, $Hf_{0.5}Zr_{0.5}O_2$
ALD	Atomic layer deposition
RIE	Reactive ion etching
RTA	Rapid thermal annealing
ECC	Error correction code
BGA	Ball grid array
LVDS	Low-voltage differential signaling
TIA	Transimpedance amplifier
DAC	Digital-to-analog converter
ADC	Analog-to-digital converter
GIPS	Giga instructions per second
IPC	Instructions per cycle
HTOL	High temperature operating life test
HTSL	High temperature storage life test
HAST	Highly accelerated stress test
ESD	Electrostatic discharge
HBM	Human body model (ESD test)
CDM	Charged device model (ESD test)
BIST	Built-in self-test

Term	Definition
DFT	Design for test

Part XIV
Appendix C: Licensing

F Document License

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G Technology License

The technology described in this document is released as a **public, patent-free engineering specification**.

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